Differential Equations in Geophysical Fluid Dynamics

XXII. Wave of quasi-geostrophic equation

Jang-Geun Choi

Center for Ocean Engineering
University of New Hampshire

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Recap

Govering equations of the quasi-geostrophic model are given by

$O(1)$ balances (predominant geostrophic balance):

$$\bar{v}_0 = \frac{g}{f_0}\frac{\partial \eta}{\partial x} \quad \text{(1a)} \qquad \bar{u}_0 = -\frac{g}{f_0}\frac{\partial \eta}{\partial y} \quad \text{(1b)} \qquad \frac{\partial \bar{u}_0}{\partial x} + \frac{\partial \bar{v}_0}{\partial y} = 0 \quad \text{(1c)}$$

$O(Ro)$ balances (small perturbations):

$$\bar{v}_1 = \frac{1}{f_0}\left(\frac{\partial \bar{u}_0}{\partial t} + \bar{u}_0\frac{\partial \bar{u}_0}{\partial x} + \bar{v}_0\frac{\partial \bar{u}_0}{\partial y} - \beta y \bar{v}_0\right) \qquad \text{(2a)}$$

$$\bar{u}_1 = -\frac{1}{f_0}\left(\frac{\partial \bar{v}_0}{\partial t} + \bar{u}_0\frac{\partial \bar{v}_0}{\partial x} + \bar{v}_0\frac{\partial \bar{v}_0}{\partial y} + \beta y \bar{u}_0\right) \qquad \text{(2b)}$$

$$\frac{\partial \eta}{\partial t} + \frac{\partial (h\bar{u}_1)}{\partial x} + \frac{\partial (h\bar{v}_1)}{\partial y} + \frac{\partial (\eta\bar{u}_0)}{\partial x} + \frac{\partial (\eta\bar{v}_0)}{\partial y} = 0 \qquad \text{(2c)}$$

Recap

Those can be rewritten as the vorticity conservation equation given by

$$\frac{\partial q_0}{\partial t} + \bar{u}_0 \frac{\partial q_0}{\partial x} + \bar{v}_0 \frac{\partial q_0}{\partial y} = 0 \tag{3a}$$

$$q_0 = \xi_0 + f - \frac{f_0}{h}\eta \tag{3b}$$

Relative vorticity of geostrophic current

$\nabla^2 \eta$

$$\xi_0 = \frac{\partial \bar{v}_0}{\partial x} - \frac{\partial \bar{u}_0}{\partial x} = \frac{g}{f_0}\left(\frac{\partial^2 \eta}{\partial x^2} + \frac{\partial^2 \eta}{\partial y^2}\right) \tag{3c}$$

Using stream function, $\bar{u}_0 = -\partial\psi/\partial y$ and $\bar{v}_0 = \partial\psi/\partial x$, it can be written as

$$\frac{\partial q_0}{\partial t} + J(\psi, q_0) = 0 \tag{4a}$$

Jacobian operator

$$J(\psi, q_0) = \frac{\partial \psi}{\partial x}\frac{\partial q_0}{\partial y} - \frac{\partial \psi}{\partial y}\frac{\partial q_0}{\partial x} \tag{4b}$$

$$q_0 = \nabla^2 \psi + f - \frac{1}{Rd^2}\psi \tag{4c}$$

Consideration of varying bathymetry

$$\frac{\partial \eta}{\partial t} + \frac{\partial((h+\eta)\bar{u})}{\partial x} + \frac{\partial((h+\eta)\bar{v})}{\partial y} = 0 \tag{5}$$

Let us decompose not only velocity component ($\bar{u} = \bar{u}_0 + \bar{u}_1$ and $\bar{v} = \bar{v}_0 + \bar{v}_1$) but also water depth $h = h_0 + h'(x, y)$ where h_0 and h' indicate constant and variation components of h.

$O(Ro)$

$$\begin{aligned}
&\frac{\partial \eta}{\partial t} + h_0\left(\frac{\partial \bar{u}_0}{\partial x} + \frac{\partial \bar{v}_0}{\partial y}\right) + h_0\left(\frac{\partial \bar{u}_1}{\partial x} + \frac{\partial \bar{v}_1}{\partial y}\right) \qquad O\left(Ro\frac{H'}{h_0}\right) \\
&+ \left(\frac{\partial (h'\bar{u}_0)}{\partial x} + \frac{\partial (h'\bar{v}_0)}{\partial y}\right) + \left(\frac{\partial (h'\bar{u}_1)}{\partial x} + \frac{\partial (h'\bar{v}_1)}{\partial y}\right) \\
&O\left(\frac{H'}{h_0}\right) + \left(\frac{\partial (\eta\bar{u}_0)}{\partial x} + \frac{\partial (\eta\bar{v}_0)}{\partial y}\right) + \left(\frac{\partial (\eta\bar{u}_1)}{\partial x} + \frac{\partial (\eta\bar{v}_1)}{\partial y}\right) = 0
\end{aligned} \tag{6}$$

Consideration of varying bathymetry

What if $Ro \ll H'/h_0$? Note that we already assumed $Ro \ll 1$. So, if $H'/h_0 \approx 1$, zeroth order balance becomes

$$\frac{\partial(h\bar{u}_0)}{\partial x} + \frac{\partial(h\bar{v}_0)}{\partial y} = 0 \tag{7}$$

that can be written as

$$\frac{\partial h}{\partial t} + \bar{u}_0 \frac{\partial h}{\partial x} + \bar{v}_0 \frac{\partial h}{\partial y} = 0 \tag{8}$$

$$\left(\because \frac{\partial h}{\partial t} = 0, \quad \frac{\partial \bar{u}_0}{\partial x} + \frac{\partial \bar{v}_0}{\partial y} = 0 \right)$$

that is equivalent to the **Taylor–Proudman theorem**!

Consideration of varying bathymetry

Quasi-geostrophic model chooses $H'/h_0 \approx Ro$. In this case, $O(Ro)$ balance of the continuity equation is given by

$$\frac{\partial \eta}{\partial t} + \frac{\partial((h' + \eta)\bar{u}_0)}{\partial x} + \frac{\partial((h' + \eta)\bar{v}_0)}{\partial y} + h_0 \left(\frac{\partial \bar{u}_1}{\partial x} + \frac{\partial \bar{v}_1}{\partial y} \right) = 0 \quad (9)$$

Note that there is no difference in momentum equations. Vorticity equation using (1), (2), and (9) is given by

$$\frac{\partial q_0}{\partial t} + \bar{u}_0 \frac{\partial q_0}{\partial x} + \bar{v}_0 \frac{\partial q_0}{\partial y} = 0 \quad (10a)$$

$$q_0 = \xi_0 + f - \frac{f_0}{h_0} (h + \eta) \quad (10b)$$

Planetary and topographic β-effect

$$\frac{\partial q_0}{\partial t} + \bar{u}_0 \frac{\partial q_0}{\partial x} + \bar{v}_0 \frac{\partial q_0}{\partial y} = 0 \tag{11a}$$

$$q_0 = \xi_0 + f - \frac{f_0}{h_0}(h + \eta) \tag{11b}$$

In case of constant depth with β-plane:

$$q_0 = \xi + \beta y - \frac{f_0}{h_0}\eta \tag{12}$$

In case of f-plane with varying topography $h = \alpha y + h_0$:

$$q_0 = \xi + \frac{\alpha f_0}{h_0} y - \frac{f_0}{h_0}\eta \tag{13}$$

So, they can do same role!

Wave-like solution to quasi-geostrophic model

Let us consider quasi-geostrophic equations written by the stream function:

$$\frac{\partial q_0}{\partial t} + \underbrace{\frac{\partial \psi}{\partial x}\frac{\partial q_0}{\partial y} - \frac{\partial \psi}{\partial y}\frac{\partial q_0}{\partial x}}_{J(\psi,\, q_0)} = 0 \tag{14a}$$

$$q_0 = \nabla^2 \psi + f - \frac{1}{Rd^2}\psi \tag{14b}$$

that can be rewritten by

$$\frac{\partial}{\partial t}\left(\nabla^2\psi - \frac{1}{Rd^2}\psi\right) + J\left(\psi, \nabla^2\psi - \frac{1}{Rd^2}\psi\right) + \beta\frac{\partial \psi}{\partial x} = 0 \tag{15}$$

Once "**monochromatic**" wave-like solution ($\psi = \psi_0 e^{i(kx+ly-wt)}$) is assumed, we can see

$$J\left(\psi, \nabla^2\psi - \frac{1}{Rd^2}\psi\right) = 0 \tag{16}$$

so it becomes equation of the linear Rossby wave perturbed by inertia!

Role of nonlinearity

Let us consider wave-like solution with two differen modes:

$$\psi = A_1 e^{\underbrace{i(k_1 x + l_1 y - w_1 t)}_{\equiv \theta_1}} + A_2 e^{\underbrace{i(k_2 x + l_2 y - w_2 t)}_{\equiv \theta_2}} \tag{17}$$

In this case, the Jacobian operator is non-zero:

$$J\left(\psi, \nabla^2\psi - \frac{1}{Rd^2}\psi\right) = A_1 A_2 (k_1 l_2 - k_2 l_1)({K_2}^2 - {K_1}^2) e^{\theta_1 + \theta_2} \tag{18}$$

where $K_n = \sqrt{{k_n}^2 + {l_n}^2}$.

Because of its nonlinearity, principle of superposition does not work!

Role of nonlinearity

Predominant (Big)!

$$\underbrace{\frac{\partial}{\partial t}\left(\nabla^2\psi - \frac{1}{Rd^2}\psi\right)}_{1/(\beta LT)} + \underbrace{J\left(\psi, \nabla^2\psi - \frac{1}{Rd^2}\psi\right)}_{\Psi/(\beta L^3)\text{: small}} + \underbrace{\beta\frac{\partial\psi}{\partial x}}_{1} = 0 \quad (19)$$

Choosing $1/(\beta LT)$ and $\Psi/(\beta L^3) \ll 1$ and applying perturbation method yields below.

$O(1)$ balance:

$$\frac{\partial}{\partial t}\left(\nabla^2\psi_0 - \frac{1}{Rd^2}\psi_0\right) + \beta\frac{\partial\psi_0}{\partial x} = 0 \quad (20)$$

$O(\Psi/(\beta L^3))$ balance:

$$\frac{\partial}{\partial t}\left(\nabla^2\psi_1 - \frac{1}{Rd^2}\psi_1\right) + \beta\frac{\partial\psi_1}{\partial x} = -J\left(\psi_0, \nabla^2\psi_0 - \frac{1}{Rd^2}\psi_0\right) \quad (21)$$

Role of nonlinearity

Solution to (21) is given by

$$\psi_1 = \frac{A_1 A_2 (k_1 l_2 - k_2 l_1)({K_2}^2 - {K_1}^2)}{i(w_3({k_3}^2 + {l_3}^2 + Rd^{-2}) + \beta k_3)} e^{\theta_3} \tag{22}$$

where $\theta_3 = i(k_3 x + l_3 y - w_3 t)$, $k_3 = k_1 + k_2$, $l_3 = l_1 + l_2$, and $w_3 = w_1 + w_2$, so it is new third mode depending on the original two modes.

If $w_3({k_3}^2 + {l_3}^2 + Rd^{-2}) + \beta k_3) = 0$,[1] the solution becomes

$$\psi_1 = -\frac{A_1 A_2 (k_1 l_2 - k_2 l_1)({K_2}^2 - {K_1}^2)}{{k_3}^2 + {l_3}^2 + Rd^{-2}} t e^{\theta_3}. \tag{23}$$

It grows as time!

that was discussed by (Longuet-Higgins and Gill, 1967).

[1] Note that this is dispersion relation of the Rossby wave perturbed by inertia

References I

Longuet-Higgins, Michael Selwyn and Adrian Edmund Gill (1967). "Resonant interactions between planetary waves". In: *Proceedings of the Royal Society of London. Series A. Mathematical and Physical Sciences* 299.1456, pp. 120–144.